

Validating the techno-economic and environmental sustainability of solar PV technology in Nigeria using RETScreen Experts to assess its viability

Abdulhameed Babatunde Owolabi, Benyoh Emmanuel Kigha Nsafon, Jong Wook Roh, Dongjun Suh, Jeung-Soo Huh*

Institute for Global Climate Change and Energy, Department of Climate Change Graduate School, Kyungpook National University, Daegu 41566, South Korea



ARTICLE INFO

Keywords:

RETScreen Expert
Solar photovoltaic
Feasibility analysis
Greenhouse gas
Emission reduction
Payback period

ABSTRACT

This paper presents a solar energy road map to attract solar energy investors to invest on clean energy technology from the underutilized and abundant solar energy in the Northeastern Nigeria to help reduce the effect of global warming caused by the burning of wood and also, enhance the sustainable technological development of the region that was hampered by the Boko Haram insurgency. To achieve this, RETScreen Expert software is used to validate the techno-economic and environmental sustainability of installing a grid-connected solar photovoltaic system in the region. Climatic data from the National Aeronautics and Space Administration (NASA) for the various locations and some financial parameters were used for the assessment. From the result, all the selected locations are recommended for solar photovoltaic project from all the available indices. But Yobe state is considered as the best site for the photovoltaic technology due to its highest annual solar radiation of 5.96 kWh/m²/d, clear sunny days and the highest capacity factor of 21.7% leading to the highest energy of 11,385MWh exported to the grid, the best payback period of 13.6 years and the maximum GHG emission reduction of 5,452.5 tons of CO₂ which is equivalent to 501.5 ha of forest absorbing carbon.

Introduction

Nigeria is naturally endowed and equally blessed with diverse and sustainable conventional and renewable energy sources such as solar, wind, hydropower, and biomass [1,2]. These potentials have the full capacity to provide both the rural villages and the urban cities of the country the sufficient energy needed for nationwide electrification [1]. Yet, the country has one of the lowest consumption rates of electricity per capita in Africa because of the obsolete equipment used at the electricity distribution and transmission station and her population that makes the demand superseding the electricity generation. The annual daily average solar radiation in the southern part of the country is 12.6 MJ/m²/day (3.5 kWh/m²/day) and 25.3 MJ/m²/day (7.0 kWh/m²/day) in the far north [3] with an annual average sunshine hour and annual average solar radiation of 6.25 h/day and 5.25 kWh/day respectively depending on the region [4]. It is possible to generate 1,850 × 10³ GWh of solar electricity per year in Nigeria with a solar module covering only 1% of the country's land area [5]. Despite her location on the equator which lies within sunshine belt where solar intensity is high and solar radiation is evenly well distributed, solar energy is yet to find its foot in the country's total energy mix even

though Nigeria is blessed with good and viable solar power application [3].

Nigeria is made up of thirty-six states and the federal capital territory (FCT) that are divided into six geopolitical zones to include North-east, North-west, North-central, South-east, South-west, and South-south. Each zone has high potentials for solar photovoltaic (PV) of different project sizes as shown in Fig. 1. For solar project classification, the six zones were grouped into three main solar radiation zones based on the amount of solar radiation gotten from each zone [3] as shown in Fig. 2. The research focus of this paper is based on Zone I which majorly comprise of all the six states in the Northeastern part of Nigeria to include Adamawa, Bauchi, Borno, Gombe, Taraba and Yobe with very high solar radiation incidence on the horizontal surface and high viable potential for large scale solar photovoltaic (PV) projects than the remaining two zones. The Northeastern part of the country produced the majority of all the agricultural crops that the citizen consumed. But the zone has achieved low rating of sustainable technological development goals when compared with other zones coupled with the insurgence of Boko Haram Terrorist Organization that have wreaked havoc to the region between 2014 and 2018 before the Buhari led administration were able to defeat them technically. Owing to the fact that, the majority of

* Corresponding author.

E-mail addresses: owolabiabdulhameed@gmail.com (A.B. Owolabi), jshuh@knu.ac.kr (J.-S. Huh).

<https://doi.org/10.1016/j.seta.2019.100542>

Received 2 July 2019; Accepted 12 September 2019

2213-1388/ © 2019 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

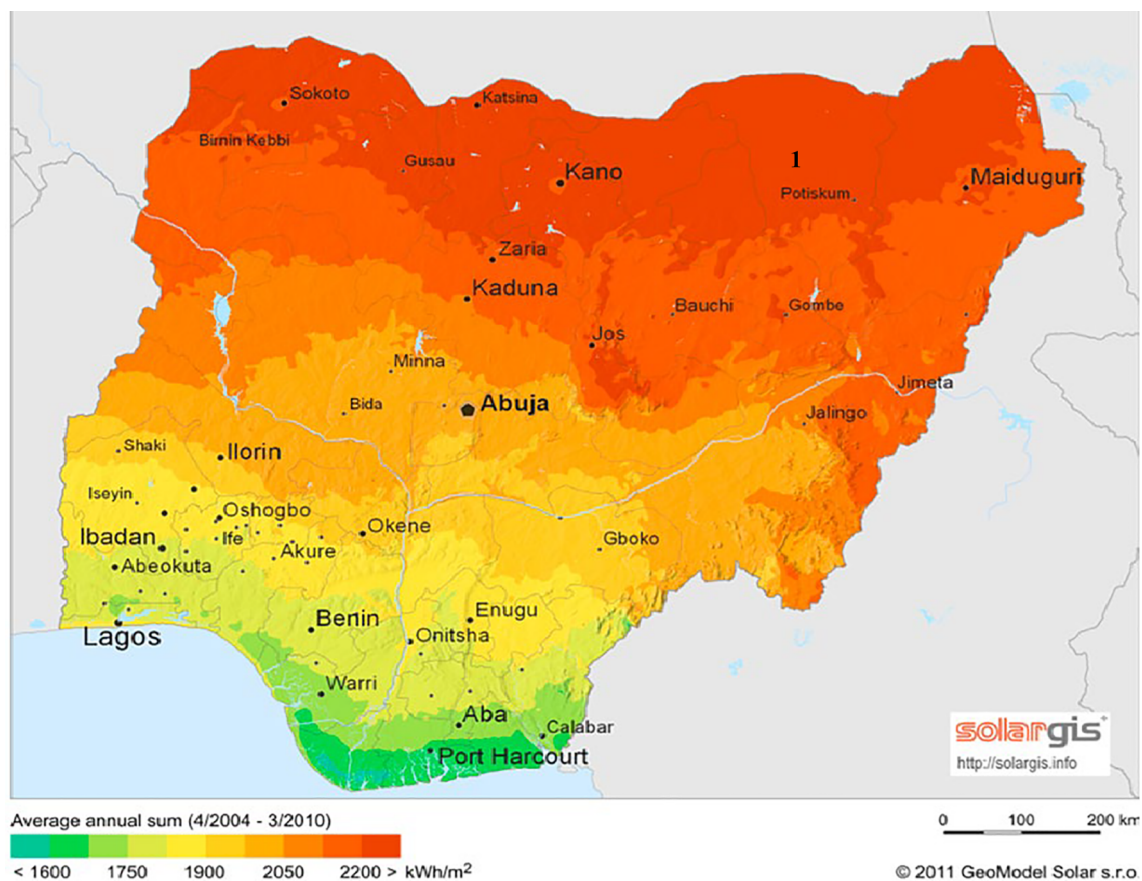


Fig. 1. Solar radiation map of Nigeria.¹ Yobe State is considered as the best site for the photo-voltaic technology due to its highest annual solar radiation, clear sunny days and the highest capacity factor.

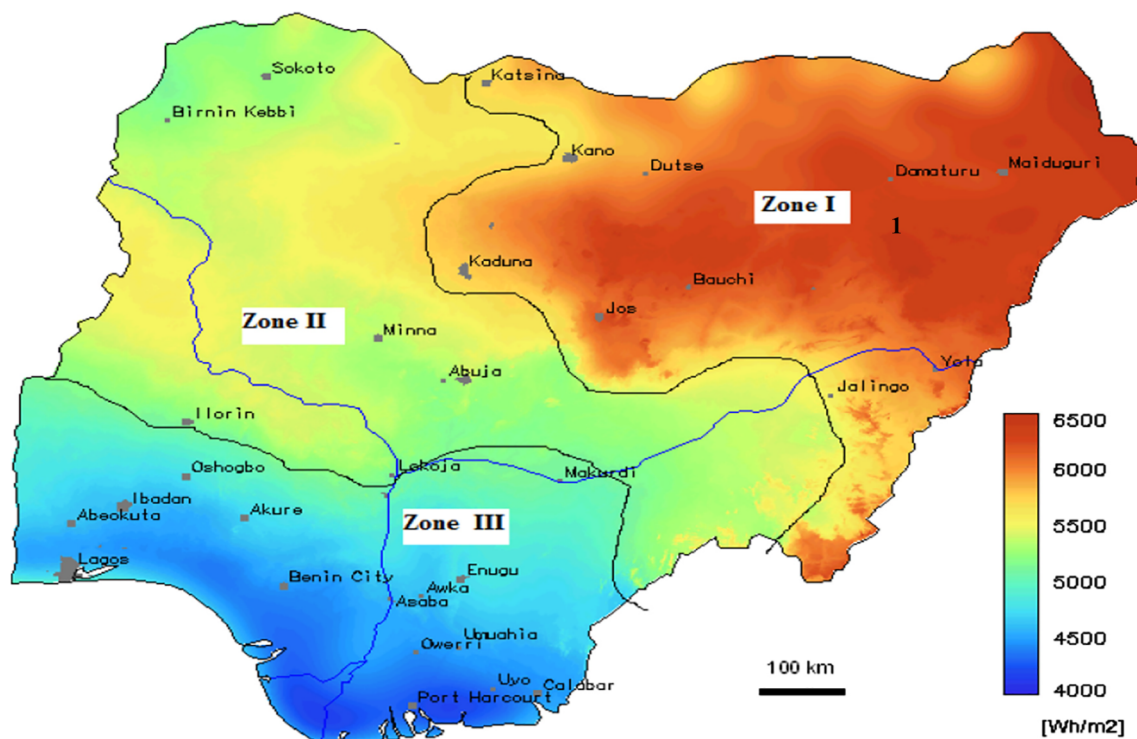


Fig. 2. Map of solar radiation zone in Nigeria.¹ Zone I comprises majorly the Northeastern parts of Nigeria with very high solar radiation incidence on the horizontal surface and high viable potential for large scale solar photo-voltaic.

the people living in the region are low income subsistence farmers that don't have access to electricity for their day-to-day activities in order to enhance the sustainable development of the region. Hence, they resorted to the use of firewood to supplement the high shortage of electricity in the region which in turn contributed to the effect of global warming as a result of the high deposit of carbon produced from the incomplete combustion of the wood in the atmosphere.

Fuelwood and charcoal are mainly used by economically and socially less privileged people [2] with over 60% of Nigerians living in the rural areas having a consumption rate of 50 million metric tons [5] most especially in the north east. This gives a total of 350,000 ha of forest and natural vegetation that is annually lost in Nigeria. Nationwide demand for fuelwood was estimated to be 39 million tons in the year 2000 alone in which 95% of the total fuelwood usage was from households, industrial and agricultural activities [5]. The majority of the fuelwood consumers are not aware of the environmental consequences of excessive extraction [2]. Sourcing fuelwood in lieu of clean energy in the Northeastern Nigeria is for domestic and commercial use and it is a major cause of desertification in the northern part of the country with the rate of deforestation equivalent to 3.6% of the present forest and woodlands. Whereas, reforestation is 10% of the deforestation rate which exceeds the replenishment rate through various afforestation programs. Hence, the need to provide and invest on clean energy technology from the underutilized abundant solar energy in the zone for sustainable development is essential. This will supplement for the unclean energy that is used by reducing the effect of global warming caused by the burning of wood in the zone and also, enhance the technological development of the region that is been hampered and exacerbate by the Boko Haram insurgency. The Federal Government of Nigeria cannot provide clean energy to approximately 20 million populations in the northeast part of the country due to some financial and economic constraints but they can attract solar energy investors from international organizations and donor countries through a solar energy road map that includes a comprehensive feasibility analysis on the technical, economic, risk, sensitivity and environmental impact on solar PV technology in the region. And this can be achieved by the use of RETScreen Expert software developed by the government of Canada. RETScreen Expert as a clean energy management software assessing tool is used to validate the techno-economic and environmental sustainability of installing a grid-connected solar photovoltaic system in the six regions and use the feasibility analysis result to convince and attract solar energy investors to invest in solar energy projects in the region. A solar PV technology of 6 MW installed capacity is considered for this paper for each of the region to include the design parameters needed for installing a solar PV technology. This will only serve as a template because the solar radiations in the region has the capacity to produce solar PV technology of any capacity with high Return of Investment (ROI) and high emission reduction.

Reviews on solar PV technology with RETScreen

Another technology that can reduce the dependence on gas power plants, which are the primary sources of CO₂ emissions causing global warming, is solar PV technology [6]. Many researchers had worked on solar PV using RETScreen and other related software like HOMER (Hybrid Optimization Model for Electric Renewable) outside Nigeria to assess the viability and government regulatory structures involved in implementing both off the grid and on grid electrification. Rashwan et al. [6] use the software to perform an economic and environmental feasibility study of switching the electrical grid into the renewable energy of a small building located in Saudi Arabia. The research evaluated the PV power plant and based the feasibility study on the new electricity tariffs for the government and commercial sectors at 4.0 and 8.0 cents/kWh when compared with the same power capability of 12 kW on the international photovoltaic project model. Shafiqur et al. [7] use RETScreen for the feasibility analysis of constructing 10 MW

installed capacity grid-connected photovoltaic power plants in Saudi Arabia by presenting the technical, environmental and economic aspect for selection of a viable site. The meteorological data like global solar radiation, sunshine duration, dry bulb temperature, and relative humidity were used to conduct the feasibility analysis in terms of energy production, greenhouse gas (GHG) emissions, and financial parameters. The study selected one photovoltaic module after the analysis.

Also, Amir et al. [8] used the software to model the solar photovoltaic systems in order to fulfill Pakistan's energy demand for the domestic sector. The research compared the role of solar irradiance and load correlation with respect to ambient conditions and observed the solar fraction by evaluating the model on the basis of the Net Present Value (NPV), Internal Rate of Return (IRR), payback periods and GHG emissions reduction. The results of the study indicate the implementation of a 5 kW stand-alone solar PV system with 0.6–0.7tCO₂ reductions in GHG emissions in different regions of Pakistan under prevailing climatic conditions. Thevenard et al. [9] assessed the potential for renewable energy projects. The study guides the user in the design of their systems, by providing initial estimates of the array, battery, or pump size. From their research, it was discovered that by changing a few of the system's parameters, users are able to quickly screen the most effective technology and system size depending on load, climatic conditions, and season. In Iran, the environmental, technical and financial feasibility study of solar power plants was done with the software by Alireza and Taraneh [10]. The study developed new electricity tariffs based on three scenarios from the RETScreen. Ayushi and Vivek [11] use RETScreen to determine the financial and environmental impact of a solar PV system for a village in India. The energy demand of the photovoltaic system was 587.7 kW according to the electrical energy demand of the village, the capital cost, payback period, discount rate and internal rate of return of the system.

Most of the Nigerian researchers that worked on RETScreen used it to address the economic and feasibility analysis of either hydropower or wind energy in some parts of the country [12–15]. And those that worked on the viability and feasibility of solar PV project in Nigeria to assess its technical and economic performance either used HOMER software or other life assessment mechanism [15–19]. Based on the research done on this paper, less research had been done within Nigeria using RETScreen to assess the sustainability of a solar PV system in some major cities of the country. Only Ahmed and Gidado [20] evaluated the potential of solar PV systems using the TRNSYS simulation environment and RETScreen simulation tool to establish the performance data for a range of solar PV products and locations. The result of the study shows that the PV systems have significant potential as stand-alone applications in buildings and other applications. Also, it can be deduced that researchers across the globe mostly worked on the technical, economic and environmental impact of solar PV using RETScreen software without considering the risk and sensitivity analysis which are vital tools in project decision making where the uncertainty of any project is analyzed by predicting the future return of the renewable energy project for sustainability.

Only few researchers like [21] included the sensitivity and risk analysis in their work when analyzing the photovoltaic solar plant in Chile. They considered the variation of $\pm 30\%$ for five parameters and their result showed that the initial cost and the electricity export rate significantly influence the project outcome. Also, [22] used it to analyze the costs and benefits of large-scale solar photovoltaic power production in Abu Dhabi, United Arab Emirates (UAE). Their result showed that the high initial costs and low expected price for electricity generated are driving reasons why photovoltaic systems are not being implemented in Abu Dhabi. Finally, [23] used it to determine the potential and viability of grid-connected solar PV system in Bangladesh. It was discovered from their result that the sensitivity analysis done on NPV varied significantly with the variation of initial cost and electricity export rate. They concluded that the sustainability of any renewable energy project can be substantiated through the use of RETScreen software.

Retscreen overview

RETScreen International is an innovative and unique renewable energy awareness, decision-support and capacity-building tool. It is developed and maintained by the Government of Canada through the Natural Resources Canada's CANMET Energy Diversification Research Laboratory (CEDRL) with center in Varennes, Quebec [8,9,24]. The software is a contribution from over 307 network of experts including industry, government and academia with collaborations from organizations like the National Aeronautics & Space Administration (NASA), the Renewable Energy and Energy Efficiency Partnership (REEEP), the United Nations Environment Program (UNEP), the Global Environment Facility (GEF), World Bank's Prototype Carbon Fund (PCF), the Energy + Environment Foundation and Leonardo Energy Initiative [24]. Aside the latest version of the software that is known as RETScreen Expert, all other versions are readily available for public use free with no cost implication for the feasibility analysis of clean energy projects, including energy-efficient technologies and renewable energy projects such as solar photovoltaic, wind energy and hydro projects [25]. The soft is in 36 languages covering more than two-thirds of the world's population in over 222 countries and territories around the world [9,24,26].

The software can be used for feasibility assessment studies of energy projects, performance evaluation of new and retrofits projects to include monitoring and evaluation of existing projects. Users can select each technology project according to the purpose of their feasibility or performance evaluation study and each technology project has a standard procedure with the same five-step analysis of the energy model to include cost analysis, greenhouse gas (GHG) analysis, financial summary, and sensitivity and risk analyses. The software helps decision-makers in evaluating the potential of certain energy project for implementation quickly and inexpensively. This is done by significantly reducing the costs and time associated with identifying and assessing potential energy projects that arise at the pre-feasibility, feasibility, development, and engineering stages [8,24]. The RETScreen International photovoltaic project model is used worldwide to easily calculate the energy production, life-cycle costs and greenhouse gas emissions reduction for three basic PV applications: on-grid; off-grid; and water pumping [10]. The latest version of the software known as a RETScreen Expert is used for this study.

RETScreen Expert worksheet

RETScreen software is used to determine if a renewable energy project makes a financial sense with an analysis capable of covering an entire project life cycle. It has three major analysis tools to include the Benchmark analysis, Feasibility analysis, and Performance analysis [24,26].

Benchmark analysis workflow

The benchmark worksheet allows the user to quickly gauge a facility's energy performance by establishing a reference climate conditions at a facility site for any location on earth, and compare the energy performance of various types of reference facilities with the estimated or measured annual energy consumption of the same facility [24,26]. The location worksheet and facility worksheet are the two worksheets under the benchmark

Feasibility analysis workflow

The feasibility analysis worksheet is where a comprehensive and detailed analysis is done. The analysis permits users to model any clean energy project by conducting a five-step standard analysis to include the energy analysis, cost analysis, emission analysis, financial analysis, and sensitivity/risk analysis [24]. The worksheet has the benchmark, product, project, hydrology and climate databases, as well as links to worldwide energy resource maps that are fully integrated [26] with an

extensive database of generic clean energy project templates as well as specific case studies.

Performance analysis workflow

The performance analysis worksheet allows a user to monitor, analyze, and report key energy performance data to facility operators, managers and senior decision-makers, including the facility's actual energy performance versus predicted performance. The worksheet uses advance regression and predictive models taking into account normalized energy performance based on varying parameters like weather data gotten from NASA to track a facility actual energy performance against its predicted performance [24,26].

Solar photovoltaic system

Solar PV system or solar power system is one of the renewable energy systems which use PV modules to convert sunlight into electricity. The electricity generated can be used directly, feed back to the grid or store in a battery. The solar PV system is a very reliable and clean source of electricity that suits a wide range of applications such as residence, industry, and agriculture [27].

Solar PV components

The major components for solar PV system are solar panel, solar charge controller, battery bank, inverter, auxiliary energy sources and loads. They are selected according to the project, site location and applications [6,9,10,19,27,28].

Solar panel

The solar panel is the fundamental energy conversion component of PV systems [29] that converts the solar energy gotten from the sun into direct current (DC) electricity [27] with an average life span between 20 and 30 years [18]. Solar panel usage ranges from aerospace industries, electric vehicles, communication equipment, irrigation, aquaculture, water pumping, and water heating. Due to fast technological development and increasing demands, the cost of PV panels has tremendously decreased [7]. PV cell technologies can be classified into six groups to include monocrystalline silicon, polycrystalline silicon, amorphous silicon, cadmium telluride (CdTe), copper indium gallium selenide (CIGS), and polymer & organic PV.

Solar charge controller

The solar charge controller is used to regulate the flow of current electricity from the photovoltaic panels into the battery by prolonging the life span of the battery and preventing it from overcharging and over-discharging when in use [10,27]. Its main function is to be able to determine whether the power generated by the solar panels is needed for the required load to run low voltage equipment and lighting or it should charge the deep-cycle backup batteries for later usage [7].

Solar battery

The battery stores energy in order to supply electrical appliances when there is a demand. It usually serves as a backup when the solar intensity is very low and the amount of solar energy generated from the panel can't be enough to carry the load. For an off-grid PV system that represents stand-alone systems, a battery is needed as a backup with or without an additional generating set. In the case of an on-grid-connected model, the use of a battery as stand by is not necessary because the energy available to the grid is what is produced by the array but in some cases, it can be useful [9]. The most commonly used battery for off-grid solar PV is the lead acid with a nominal voltage of 2VDC because of its durability as a result of non-leakage of the electrolyte from the battery's terminal [6,10].

Solar inverter

The inverter is the heart of every solar power system because it is the key to the safety, efficiency, and reliability of any system. It is an electronic device that converts direct current (DC) output of PV panels into clean alternating current (AC) [27]. They are designed to handle high voltage and current in a very ash weather conditions like high temperature, water, and voltage fluctuation and usually come in two models; On-grids and Off-grids [6,10]. The ingress protection (IP) rating of an inverter defines the ability of an inverter seal to prevent it from foreign objects like dust and water. The inverter IP rating and the inverter efficiency are the two most important factors that determine the life span of the inverter.

Design parameters for a photovoltaic solar system

The design methodology, parameters and the technical specifications of the photovoltaic system that is considered for the solar PV project in the northeastern part of Nigeria is discussed in this chapter. This include the power generation factor, energy demand, solar PV energy required, PV module sizing, inverter sizing and the battery sizing [28,30,31]. A 6 MW photovoltaic solar system is proposed for the project to power a cluster of 10,000 homes in each location consisting of 6,000 flats of two-bedrooms and 4,000 flats of three-bedrooms.

Power generating factor

Panel generation factor (PGF) is an important factor considered in sizing of solar photovoltaic technology on the basis of total watt peak rating of the system. The PGF for Nigeria is calculated as 3.41 after using the below equation [27].

$$\text{Panel Generation Factor} = \frac{\text{Solar irradiance} \times \text{Sunshine hours}}{\text{Standard test conditions irradiance}} \quad (1)$$

Energy demand

In designing a solar PV system, the first step is to calculate the total power and energy consumption of all loads that will be supplied by the system for each house and sum it up to have the total energy demand for the project [27,30]. According to [32], the power estimate for an average 3-bedroom and a 2-bedroom apartment in Nigeria are 2511Wh/day and 956Wh/day respectively. This gives a total watt-hour per day of 15,780,000 which is the energy that is needed by the 10,000 bedroom clusters for each location as shown in Table 1.

Solar PV energy required

The energy that must be generated by the PV module is the total Watt-hour per day needed from the PV modules and it is calculated by multiplying the peak energy requirement which is the total Watt-hour per day for the 10,000 houses by the energy lost in the system.

- Peak energy requirement = 15,780,000
- Energy lost in the system = 1.3 [27,28]
- Energy required from PV modules = $15,780,000 \times 1.3 = 20,514,000$ Wh/day

Table 1

Total watt-hour per day for 10,000 cluster houses.

Bedroom types	Daily power consumption (Wh/day)	No of houses	Total watts hour per day (Wh/day)
3 bedroom	2511	4,000	10,044,000
2 bedroom	956	6,000	5,736,000
Total	3467	10,000	15,780,000

PV module sizing

To calculate the size of the photovoltaic modules that is required, the first thing to estimate is the total Watt-peak rating needed for the PV modules which is given as 6,015,835 Watts by using the formular below

$$\text{Total Watt peak rating} = \frac{\text{Solar PV energy required}}{\text{Panel generation factor}} \quad (2)$$

This is then followed by calculating the PV module size which is the ratio of the total watt peak rating to the PV output power rating as shown in the formular below. This gives a solar module of 30,000 when considering China Sunergy mono-si-CSUN200-72M with an output power rating of 200 W.

$$\text{PV module size} = \frac{\text{Total Watt peak rating}}{\text{PV output power rating}} \quad (3)$$

The China Sunergy mono-si-CSUN200-72M with technical details given in Table 2 was considered because of its availability and cheapness in Nigerian market when compared to other solar PV modules. The module has one of the highest leading conversion efficiency with a positive tolerance offer, it is also certificated to withstand wind up to 2,400 Pascal, snow load of 5,400 Pascal and to have passed salt mist, ammonia, blowing sand and hail testing in ash environment. Also, the panel has an excellent performance under weak light conditions, good temperature coefficient for better output in high-temperature regions like Nigeria. It should be noted that with fewer PV modules, the system will not work efficiently especially during cloudy days and this will shorten the working life span of the battery. But if the number of PV modules used is more than the minimum required during project installation, it will enhance the system performance by also improving the battery life span. This is because the amount of electricity generated from a PV panel is a factor of the panel size, the panel position, the panel wattage and the climate of site location [7].

Inverter sizing

The size of an inverter used for any solar photovoltaic project is a function of the total wattage of the energy consumed and the factor of safety [31], and it is calculated as shown below.

Peak energy requirement = 15,780 kW

Factor of safety = 1.3 [31]

Inverter size = $15,780 \times 1.3 = 20,514$ kW

Two thousand five hundred and sixty four of Digital Luminous Inverter with a rating of 10 kVA/180 V is considered for this project by taking the power factor of 0.8 when converting the kVA to kW. This is

Table 2

The technical details of China sunergy mono-Si - CSUN200-72M.

S/N	Parameters	Values	Units
1	Maximum Power (P_{max})	200	W
2	Positive Power Tolerance	0–3	%
3	Open Circuit Voltage (V_{oc})	45.3	V
4	Short Circuit Current (I_{sc})	5.72	A
5	Maximum Power Voltage (V_{mpp})	37.6	V
6	Maximum Power Current (I_{mpp})	5.32	A
7	Module Efficiency	15.67	%
8	Voltage Temperature Coefficient	−0.307	%/K
9	Current Temperature Coefficient	+0.039	%/K
10	Power Temperature Coefficient	−0.423	%/K

Note 1: Electrical data relates to standard test conditions (STC): irradiance 1000 W/m²; Air Mass 1.5; cell temperature 25 °C measuring the uncertainty of power is within ± 3%. Certified in accordance with IEC61215, IEC61730-1/2.

because of its high output capacity offering robust backup performance when compared to other similar inverters and its availability in the Nigerian market. The inverter input rating having the same nominal voltage as the project battery should be higher than the total watt of the entire appliance that is used so as to handle the total peak watt requirement of the project [27,28]

Battery sizing

The Ampere-hour capacity of the deep-cycle battery for each of the two bedroom apartment is rated 12 V 469 Ah and 12 V 1230 Ah for each of the three bedroom apartment using the formular below

$$CB = \frac{\text{Daily power consumption} \times \text{Days of autonomy}}{\text{Battery efficiency} \times \text{Depth of discharge} \times \text{Battery nominal Voltage}} \quad (4)$$

where, daily power consumption is 15,780 kW, days of autonomy is 3, battery efficiency is 0.85, battery nominal voltage is 12 V and battery depth of discharge is 0.6. For solar energy project, the deep cycle battery that is specifically designed for a low energy level discharge is recommended [27]. The lead acid battery which is the building block of the energy storage system should be large enough to store sufficient energy that will operate the whole buildings or appliances at night and on cloudy days by avoiding power interruption from the solar photovoltaic system [27,30,31].

Results and discussion

This study used RETScreen Expert software to assess the viability of installing a 6 MW grid-connected solar photovoltaic system in the six northeastern states of Nigeria. The climatic data of the various locations were calculated by the RETScreen using the National Aeronautics and Space Administration (NASA) data as listed in Table 3. Yobe state has the highest annual solar radiation of 5.96 kWh/m²/d followed by Borno state and Taraba state having the least. Also, the variation of the average monthly solar radiation of the selected locations is shown in Fig. 3. A thorough feasibility analysis carried out shows the technical, economic, environmental and risk/sensitivity analysis result

Technical viability

In engineering, the technical sustainability takes into consideration the product specification upon which the efficiency and efficient utilization depend [8]. Fixed solar tracking mode is considered for the six locations with the solar photovoltaic module tilted to an angle of 15 degrees. These was chosen because the solar irradiance and the electricity exported to the grid for each location will be at the highest when the panel tracking is fixed to the base and sloped to an angle of 15 degree. The RETScreen Expert software calculated the parametric characteristics of the China Sunergy mono-si-CSUN200-72M solar module as shown in Table 4.

Table 3
Climatic data of the selected locations.

Location	Adamawa	Bauchi	Borno	Gombe	Taraba	Yobe
Latitude	9.2	10.3	11.9	10.3	8.9	11.8
Longitude	12.5	9.8	13.2	11.2	11.4	12
Elevation	172	617	323	456	227	370
Heating design temperature (°C)	18.7	17	17.4	17.9	18.8	16.1
Cooling design Temperature (°C)	35	33.9	36.8	35.2	33.8	36.5
Earth temperature amplitude (°C)	15.9	16.3	18.5	16.7	14.1	19
Average monthly solar radiation (kWh/m ² /d)	5.69	5.76	5.89	5.77	5.57	5.96

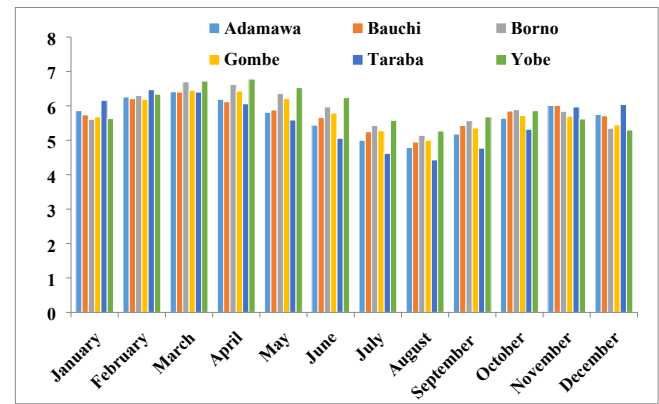


Fig. 3. Monthly variation of the daily solar radiation for the six locations.

The energy produced from any solar photovoltaic module depends on the solar radiation of the location and the number of clear sunny days [8,11] and this, in turn, affects both the annual energy exported to the grid by the panel and the capacity utilization factor (CUF) which is the ratio of electrical energy generated by solar photovoltaic plant over the year to the energy output at its rated capacity over the year [11]. The solar module in Yobe state has the highest capacity factor of 21.70% when compared with the solar modules of other locations as shown in Table 5. This is as a result of the amount of solar irradiance gotten from the state. Also, the annual energy exported to the grid varied between a minimum of 10,743 MWh in Taraba state and a maximum of 11,385 MWh in Yobe state and this affects the annual electricity exported revenue of the locations. According to the results gotten by [7] when they analyzed the feasibility study of the grid-connected 10 MW installed capacity photovoltaic power plants in Saudi Arabia. The values gotten as the annual energy exported to the grid and their capacity factor in each location in the Northeastern states of Nigeria are compatible with the acceptable values and hence, it is technically sustainable to build and operate a solar photovoltaic plants in the six locations based on the technical viability of the solar system.

Table 4
Parametric characteristics of China sunergy solar module.

Property	Value
PV technology type	mono-Si
Power capacity	6 MW
Manufacturer	China Sunergy
Model	mono-Si-CSUN200-72M
Number of units	30,000
Efficiency	15.67%
Nominal operating temperature	45 °C
Temperature coefficient	0.4%/°C
Solar collector area	38,290 m ²

Table 5
Solar photovoltaic energy summary per location.

Locations	Capacity factor	Annual energy exported	Annual electricity exported revenue
Adamawa	20.70%	10,893MWh	\$1,252,667
Bauchi	21.20%	11,120MWh	\$1,278,748
Borno	21.40%	11,256MWh	\$1,294,471
Gombe	21.00%	11,058MWh	\$1,271,702
Taraba	20.40%	10,743MWh	\$1,235,466
Yobe	21.70%	11,385MWh	\$1,309,232

Economic sustainability

In developing a solar photovoltaic project, the economic analysis is a very important analysis that is done to know if a project will be economically viable and sustainable. The financial analysis worksheet of the software contains some financial parameters to include inflation rate, discount rate, reinvestment rate, debt ratio, debt interest rate as input variables as shown in Table 6. The values used as input variables are standards gotten directly from the software [26] excluding the initial cost and the operation and maintenance cost that was manually inputted. Based on the inputted variables, the Internal Rate of Return (IRR), the Net Present Value (NPV), the annual life savings and other financial parameters were calculated by the RETScreen software as shown in Table 7. According to [8], the economic viability of a project is a measure of its NPV, IRR and the payback period. For this project, the NPV which is the difference between the present value of cash inflows and the present value of cash outflows over a period of time for the six locations were positive and this makes the project to be financially and economically feasible according to [6–8,10,11]. Also, the value of the IRR which is a measure of a project's profitability that is gotten for the six locations is higher than the required rate of return of the project [7,8]. This makes the projects in the six locations to be economically acceptable when considering the required rate of return of the project has the discount rate.

The simple payback period which is the length of time that will take to recoup the project's initial investment in the six locations is illustrated in Fig. 4. The proposed photovoltaic project in Taraba state has the longest payback period of 14.6 years followed by the one in Adamawa state with 14.4 years, while Yobe state has the shortest payback period of 13.6 years. Irrespective of the break-even year, the solar PV systems in all the location makes financial sense. But the one to be sited in Yobe state has the highest equity payback of 14.2 years while the one in Taraba state has the least with 15.7 years. This simply means that the project will yield interest in Yobe state for 10.8 years and 9.3 years in

Taraba out of the 25 years designed for the project to work. The profit that each project location will accumulate after recouping the total cost incurred on the project is shown in Fig. 5 with the project in Yobe state making \$156,402 followed by the one to be sited in Borno with \$151,477 and the one in Taraba state making the least at \$131,788 after the expiration of the projects

Emission reduction assessment

The emission analysis worksheet is used to calculate the greenhouse gas (GHG) emission reduction resulting from carrying out the solar PV installation. It is also used to calculate the revenue that may occur from the sales of the GHG reduction emission. The transmission and distribution losses of 16% [33] and GHG emission factor of 0.433tCO₂/MWh were considered as the base case electricity system in Nigeria. The software calculates the gross annual GHG emission reduction for each of the six locations by subtracting the calculated emission in the proposed case from the calculated emission in the based case as shown in Table 8. This is because the GHG emission during the project operational period is only considered and not during the life cycle of the project. The project in Yobe state has the maximum GHG emission reduction of 5,452.5 tons of CO₂ that is equivalent to 501.5 Hectares of forest absorbing carbon not cut down for the entire life span of the project if installed as shown in Fig. 6. This is then followed by the project in Borno state and the least emission reduction coming from the project in Taraba state. The Global Forest Resources Assessment (FRA), coordinated by FAO, found that the world's forest area decreased from 31.6 percent of the global land area to 30.6 percent between 1990 and 2015 [34], but the pace of loss has slowed in recent years because of development of clean energy technology for sustainable development. According to the sustainable development goals (SDGs), countries under the United Nations made a commitment to achieve sustainable development agenda by 2030 to tackle the complex challenges human faced from ending poverty and hunger and responding to climate change to building resilient communities, achieving inclusive growth and sustainably managing the Earth's natural resources. Hence this project will help in achieving the set goals if implemented.

Table 9 shows the GHG reduction revenue that is proposed to any investor in other to generate revenue by selling the GHG reduction emission through carbon trading. The GHG reduction credit rate proposed is \$10 over a 15 year period at an escalation rate of 2% annually. The software calculated the GHG reduction revenue for the six locations with the project that will be sited in Yobe state making a profit of \$5,180 for the first year followed by the project in Borno state making \$5,121 and the least profit coming from the project in Taraba state. This is due to the fact that Taraba state has the minimum amount of GHG reduction emission as compared to other states.

Sensitivity analysis

The level of uncertainty associated with the project during analysis is a measure of the level of uncertainty of the inputted variables which has effects on the level of uncertainty on the calculated financial

Table 6
Financial input variables from RETScreen Expert.

Financial parameters	
Escalation rate fuel	2%
Inflation rate	2%
Discount rate	6%
Reinvestment rate	9%
Project Life	25yrs.
Debt ratio	60%
Debt interest rate	5%
Debt term	15%
Debt payment	\$832,397/yr.
Electricity export rate	\$0.12/kWh
Electricity export escalation rate	2%
Initial cost	\$14,400,000
Operation and maintenance cost	\$300,000
Total annual cost	\$1,132,397

Table 7
Financial output variables from the RETScreen Expert.

Financial viability	Adamawa	Bauchi	Borno	Gombe	Taraba	Yobe
Internal rate of return (%)	7.9	8.4	8.6	8.2	7.6	8.9
Net Present Value (NPV) (\$)	1,716,405	2,138,776	2,393,415	2,024,675	1,437,841	2,632,454
Annual life cycle savings (\$)	134,269	167,309	187,229	158,384	112,478	205,928
Benefit-cost (B-C) ratio	1.3	1.4	1.4	1.4	1.2	1.5
Debt service charge	1.2	1.3	1.3	1.3	1.2	1.3
Energy production Cost (\$/kWh)	0.133	0.131	0.129	0.131	0.135	0.128

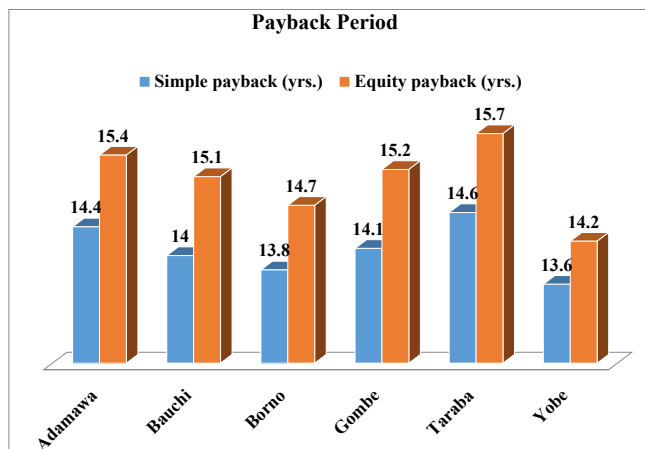


Fig. 4. The project payback period for the six locations.

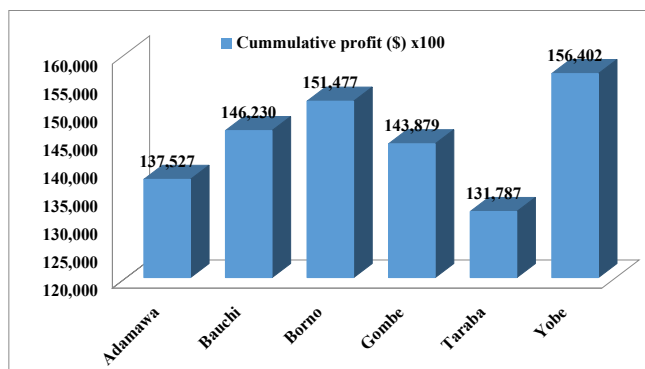


Fig. 5. The cumulative cash flow of the project after 25 years for the six locations.

Table 8
Annual GHG emission for the six locations.

Location	Based case (tCO ₂)	Proposed case (tCO ₂)	Gross annual GHG emission reduction (tCO ₂)
Adamawa	5,609.6	392.7	5,216.9
Bauchi	5,726.4	400.8	5,325.5
Borno	5,796.8	405.8	5,391.0
Gombe	5,694.8	398.6	5,296.2
Taraba	5,532.5	387.3	5,145.3
Yobe	5,862.9	410.4	5,452.5

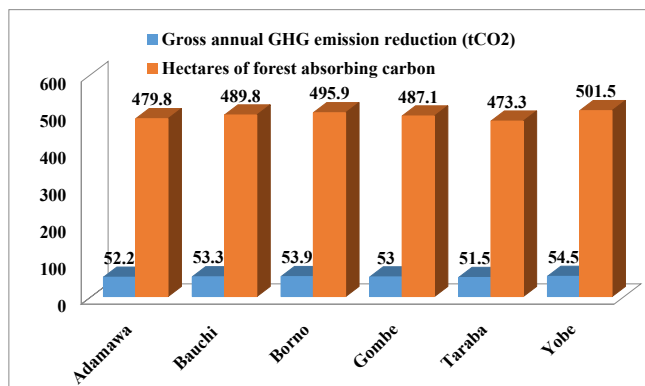


Fig. 6. Gross annual GHG emission and the corresponding barrels of crude oil not used.

variables. The sensitivity analysis worksheet is where the level of uncertainty can be reduced by considering two inputted parameters against the calculated financial variables. Only the solar photovoltaic project in Yobe state is considered for this section since it has the best technical, economic and environmental analysis when compared with other locations. The sensitivity analysis was done for the Net Present Value (NPV) of the project by varying the initial cost against the debt interest rate by $\pm 30\%$ for scenario 1 and the initial cost against the electricity exported to the grid by the same range for scenario 2 as shown in Fig. 7.

For the first case, the value of the initial cost is gotten to be \$14,400,000 and with a $\pm 30\%$. It will be \$18,720,000 and \$10,080,000 respectively. Also, the actual debt interest rate is 5.0% but with a $\pm 30\%$ it will be 6.5% and 3.5% respectively. By recalculating the NPV for the combination of initial cost and debt interest rate holding all other parameters fixed. The software indicates the values of the NPV below the threshold of zero by orange colour. With an increase of 30% to the initial cost and a decrease of 30% to the debt interest rate, the project will financially not profitable because the NPV will be $-\$482,667$. But if the initial cost is reduced by 30% and the debt interest rate is increased by 30%, the project will be financially feasible because the NPV would be \$6,197,775 which is far greater than zero. This means that the NPV is more sensitive to the initial cost than the debt interest rate for this project.

Similarly, for the second scenario, where the initial cost is considered against the electricity exported to the grid, it is observed that for the project to be financially profitable, the value of the initial cost must be reduced as against the value of the electricity exported to the grid that must increase. In both scenarios, the Net Present Value (NPV) is far more sensitive to the initial cost than to the debt interest rate and the electricity exported to the grid.

Risk analysis

Risk analysis, just like the sensitivity analysis should be done before the commencement of any solar energy project so that the level of uncertainty in a project will be known. The only difference between the two analysis is that, instead of using two parameters to reach a conclusion for a risk analysis as in the case of a sensitivity analysis, all the parameters are allowed to vary with each other during the analysis within a specific range. The energy production cost is selected as the financial indicator and the range considered is $\pm 25\%$ for all the parameters. The risk analysis is done only for the solar photovoltaic in Yobe state as it is done for the sensitivity analysis. The RETScreen software performed a Monte Carlo simulation techniques for 5000 times by recalculating the energy production cost and the result gotten is displayed as the impact graph and the distribution graph as shown in Figs. 8 and 9.

According to the impact graph, the variation on the energy production cost is as a result of the variation of the various parameters. The initial cost and the electricity exported to the grid have a great impact on the project but in the opposite direction. An increase in the initial cost increases the energy production cost while an increase in the electricity exported to the grid decreases the energy production cost. In the distribution graph, the level of risk is set at 20% by excluding the lower 10% and the upper 10% values expected from the energy production cost to exceed a 90% confidence level (P90 value). According to the result gotten, The P90 energy production cost is \$146 per megawatt-hour and this is closer to the \$128 per megawatt-hour of the energy production cost gotten in Yobe state making the project to be bankable and realistic.

Conclusion

In this study, a thorough feasibility analysis was conducted to assess the viability of installing a 6 MW grid-connected solar photovoltaic

Table 9
Annual GHG emission revenue for the six locations.

GHG reduction in revenue	Adamawa	Bauchi	Borno	Gombe	Taraba	Yobe
Net GHG reduction (tCO ₂ /yr.)	4,956	5,059	5,121	5,031	4,888	5,180
Net GHG reduction for 25yrs. (tCO ₂)	123,901	126,481	128,036	125,784	122,200	129,496
GHG reduction credit rate (\$/tCO ₂)	10	10	10	10	10	10
GHG reduction revenue (\$)	49,561	50,592	51,215	50,314	48,880	51,798
GHG reduction credit duration (yr.)	15	15	15	15	15	15
Net GHG reduction for 15yrs. (tCO ₂)	74,341	75,889	76,822	75,471	73,320	77,698

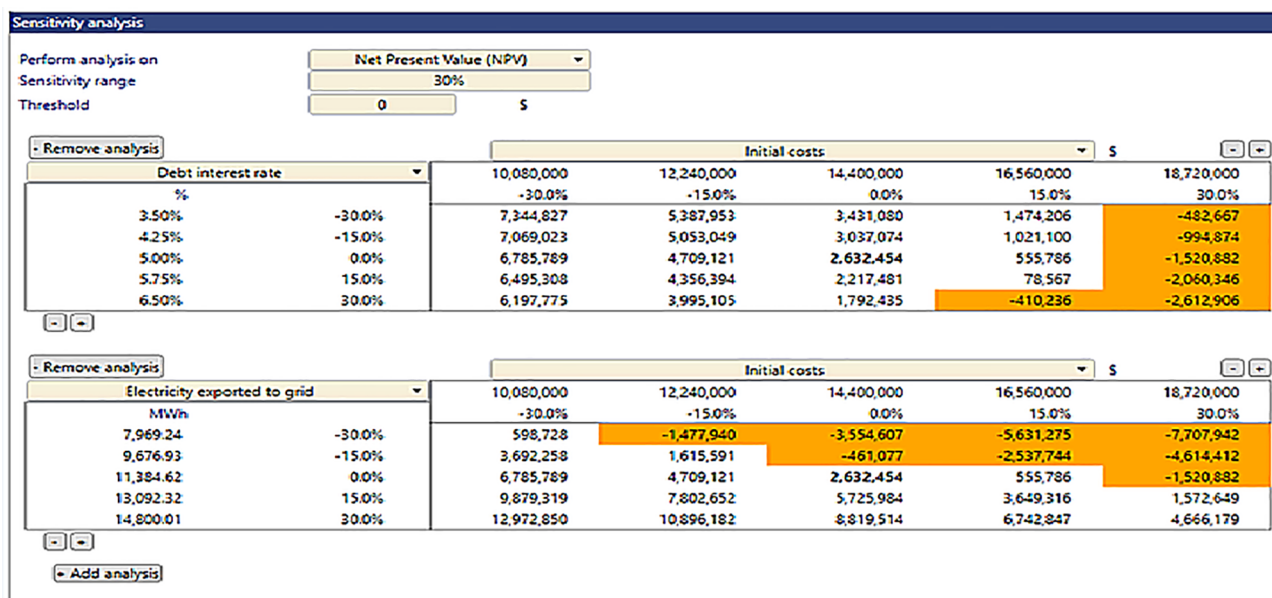


Fig. 7. Sensitivity analysis result for the solar photovoltaic project in Yobe State.

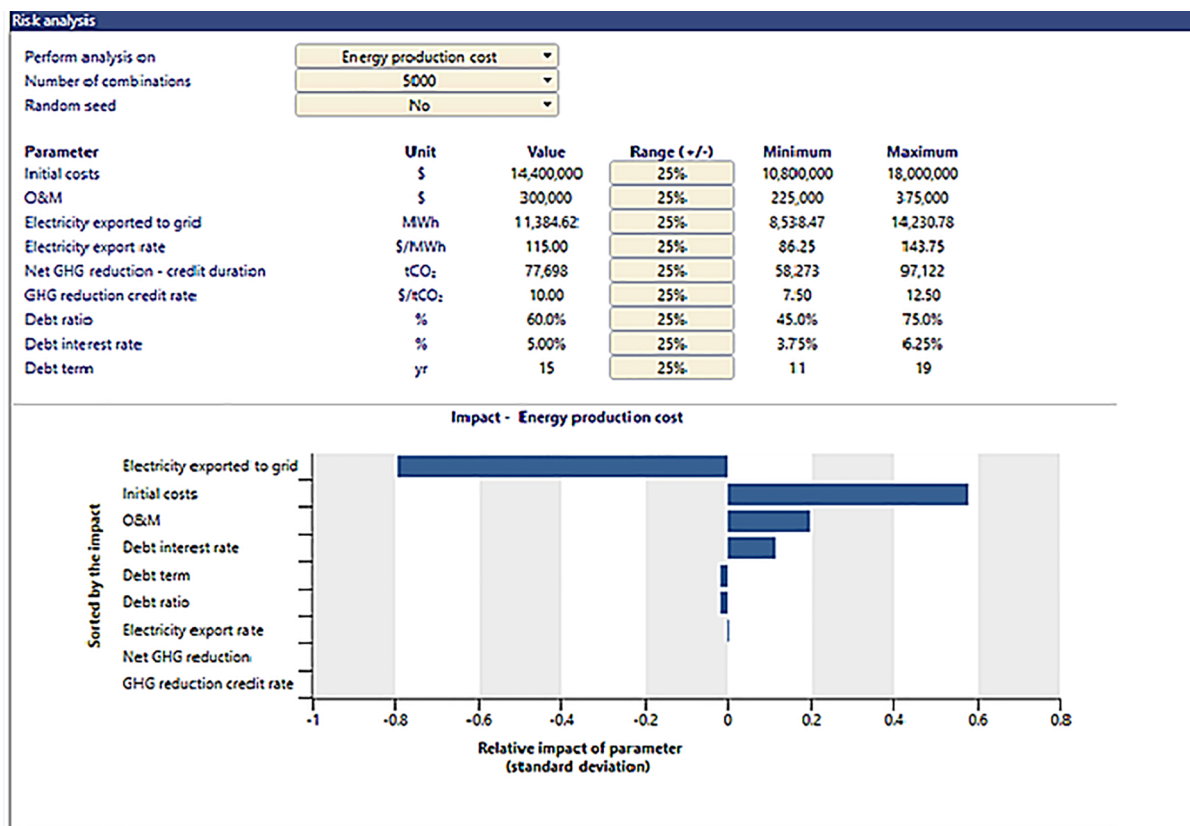


Fig. 8. Impact graph of the risk analysis result for the solar photovoltaic project in Yobe State.

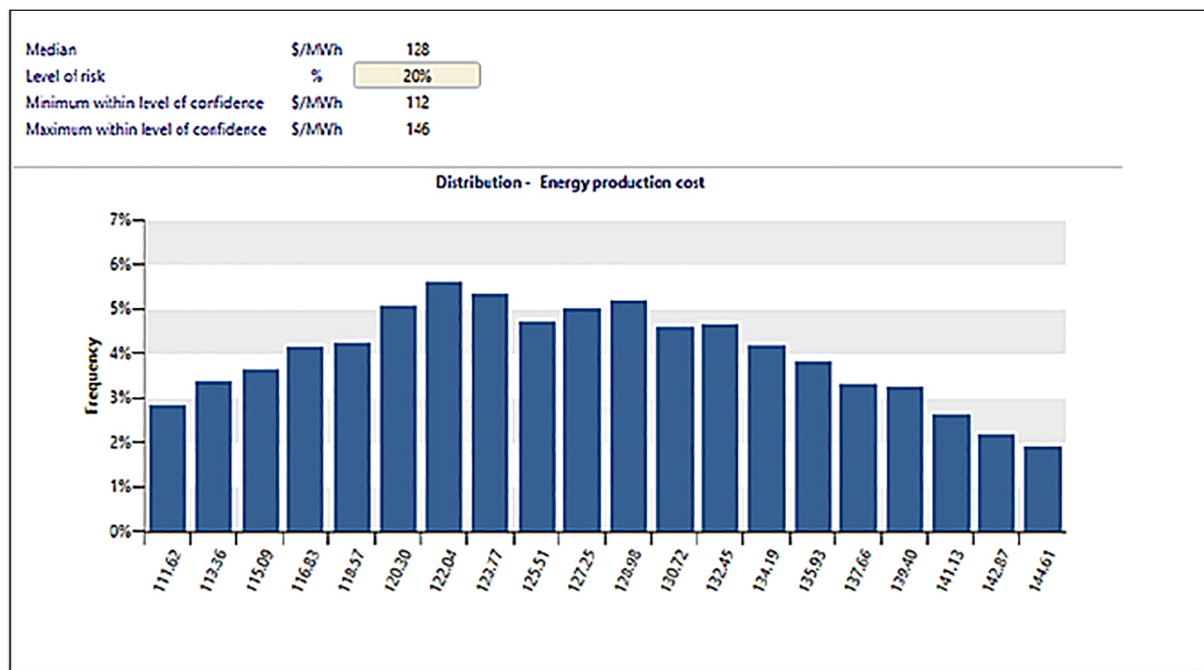


Fig. 9. Distribution graph of the risk analysis result for the solar photovoltaic project in Yobe State.

project in the six northeastern states of Nigeria by analyzing the technical, financial, risk, sensitivity and environmental impact of the project on the host community. This was done to have a solar energy road map that will attract solar energy investors from the international organizations and donor countries in order to invest in the underutilized and abundant clean energy in the Northeastern Nigeria. This will help to reduce the effect of global warming caused by the burning of wood in the zone and also, enhance the sustainable technological development of the region that was hampered and exacerbated by the Boko Haram insurgency. The Climatic data from NASA for the various locations and some financial parameters to include the reinvestment rate, inflation rate, discount rate, and debt interest rate were used for the analysis as input data inside the RETScreen Expert software. Depending on the selected location and the solar panel wattage which was the same for this work, the software accurately calculate the total amount of energy generated by the panel and exported to grid, the amount of greenhouse gas (GHG) emission that will be reduced annually, the hectares of forest absorbing carbon, the greenhouse gas emission revenue that will be generated during the life cycle of the project, the financial implication of the project to include the amount of revenue that will be generated from the sales of the energy exported to the grid and the risk involved in executing the project. All the selected locations are perfectly recommended for the installation of the solar photovoltaic project from all the available indices, but the solar project site in Yobe state is considered to be the best because of the amount of annual solar radiation and the capacity factor of the solar panel which contributes to the high energy exported to the grid. The payback period of the solar panel in Yobe state is also calculated to be the shortest. This is as a result of the high solar radiation and clear sunny days which has a greater effect on the amount of energy exported to the grid in the location. Also, the state has the maximum GHG emission reduction of 5,452.5 tons of CO₂ which is equivalent to 501.5 Hectares of forest absorbing carbon not used for the entire period of the project. From the sensitivity and risk analysis done for Yobe state, it is known that a decrease in the initial cost of the project has a great impact on the Net Present Value (NPV) and the energy production cost of the project. While an increase in the electricity exported to the grid decreases the energy production cost after setting the confidence level to 90%.

Finally, the state government in the northeastern region should not rely on the federal government hydro and gas-fired power plants that is epileptic since time immemorial because of technical, political and environmental factors, rather they should partner with the Rural Electrification Agency (REA) in searching for solar energy investors that will utilize the abundance solar irradiances and clear sunny days in their respective states. This should be done in order to provide stable electricity through the installation of the solar photovoltaic farm of different capacity in accordance with the need of each state since the region has the potential according to the analysis conducted with the RETScreen software. This will help the region to develop in an exponential rate and will drastically reduce the effect of global warming from the region as a result of the deposit of the GHGs that is emitted by the burning of woods from the zone

Acknowledgments

This work was supported by “Human Resources Program in Energy Technology” of the Korea Institute of Energy Technology Evaluation and Planning (KETEP), granted financial resource from the Ministry of Trade, Industry & Energy, Republic of Korea (No. 2019401000040).

References

- [1] Shaaban M, Petinrin JO. Renewable energy potentials in Nigeria: meeting rural energy needs. *Renewable Sustainable Energy Rev* 2014;29:72–84. <https://doi.org/10.1016/j.rser.2013.08.078>.
- [2] Mohammed YS, Mustafa MW, Bashir N, Mokhtar AS. Renewable energy resources for distributed power generation in Nigeria: a review of the potential. *Renewable Sustainable Energy Rev* 2013;22:257–68. <https://doi.org/10.1016/j.rser.2013.01.020>.
- [3] Ohunakin OS, Adaramola MS, Oyewola OM, Fagbenle RO. Solar energy applications and development in Nigeria: drivers and barriers. *Renewable Sustainable Energy Rev* 2014;32:294–301. <https://doi.org/10.1016/j.rser.2014.01.014>.
- [4] Fadare DA. Modelling of solar energy potential in Nigeria using an artificial neural network model. *Appl Energy* 2009;86:1410–22. <https://doi.org/10.1016/j.apenergy.2008.12.005>.
- [5] Sambo AS. Strategic developments in renewable energy in Nigeria. *Int Assoc Energy Econ* 2009;9–15.
- [6] Rashwan SS, Shaaban AM, Al-Suliman F. A comparative study of a small-scale solar PV power plant in Saudi Arabia. *Renewable Sustainable Energy Rev* 2017;80:313–8. <https://doi.org/10.1016/j.rser.2017.05.233>.

- [7] Rehman S, Ahmed MA, Mohamed MH, Al-Sulaiman FA. Feasibility study of the grid connected 10 MW installed capacity PV power plants in Saudi Arabia. *Renewable Sustainable Energy Rev* 2017;80:319–29. <https://doi.org/10.1016/j.rser.2017.05.218>.
- [8] Mehmood A, Shaikh FA, Waqas A. Modeling of the solar photovoltaic systems to fulfill the energy demand of the domestic sector of Pakistan using RETScreen software. *Int Conf Util Exhib Green Energy Sustainable Dev* 2014;2014:1–7.
- [9] Thevenard D, Leng G, Martel S. The retscreen model for assessing potential PV projects. *Conf Rec IEEE Photovolt Spec Conf* 2000. p. 1626–9. <https://doi.org/10.1109/PVSC.2000.916211>.
- [10] Mirzahosseini AH, Taheri T. Environmental, technical and financial feasibility study of solar power plants by RETScreen, according to the targeting of energy subsidies in Iran. *Renewable Sustainable Energy Rev* 2012;16:2806–11. <https://doi.org/10.1016/j.rser.2012.01.066>.
- [11] Khandelwal A, Shrivastava V. Viability of grid-connected solar PV system for a village of Rajasthan. *IEEE Int Conf Information, Commun Instrum Control (ICICIC)* 2017) 2018. p. 1–6. <https://doi.org/10.1109/ICOMICON.2017.8279175>.
- [12] Otugoma SO, Ogujor EA, Kuale PA. Evaluation of the small hydropower potential of river Ethiopia using the RETScreen software. *Int J Eng Res* 2016;5:1–5.
- [13] Adejumo SO, Dare AA, Ojaomo OA. Harnessing hydropower potential in Ekiti State. *Nigeria* 2015;3:319–21.
- [14] Dioha MO, Dioha JI, Olugboji OA. An assessment of Nigeria wind energy potential based on technical and financial analyses. *J Sustainable Energy* 2016;7:53–7.
- [15] Ikeagwuani I, Bamsile O, Abbasoglu SAJ. Performance comparison of PV and wind farm in four different regions of Nigeria. 2016. p. 1–16.
- [16] Okoye CO, Taylan O, Baker DK. Solar energy potentials in strategically located cities in Nigeria: review, resource assessment and PV system design. *Renewable Sustainable Energy Rev* 2016;55:550–66. <https://doi.org/10.1016/j.rser.2015.10.154>.
- [17] Adaramola MS. Viability of grid-connected solar PV energy system in Jos, Nigeria. *Int J Electr Power Energy Syst* 2014;61:64–9. <https://doi.org/10.1016/j.ijepes.2014.03.015>.
- [18] Uduak SA, Salisu RI, Udoakah YN. Electricity access in Nigeria: Viability of off-grid photovoltaic system 2013. <https://doi.org/10.1109/AFRCON.2013.6757778>.
- [19] Akinyele DO, Rayudu RK. Distributed photovoltaic power generation for energy-poor households: the Nigerian perspective. *Asia-Pacific power and energy engineering conference* 2013. <https://doi.org/10.1109/APPEEC.2013.6837165>.
- [20] Ahmed A, Gidado K. Evaluating the potential of renewable energy technologies for buildings in Nigeria. *24th Annu ARCOM Conf*. 2008. p. 1175–82.
- [21] Bustos F, Toledo A, Contreras J, Fuentes A. Sensitivity analysis of a photovoltaic solar plant in Chile. *Renewable Energy* 2016;87:145–53. <https://doi.org/10.1016/j.renene.2015.09.070>.
- [22] Harder E, Gibson JMD. The costs and benefits of large-scale solar photovoltaic power production in Abu Dhabi United Arab Emirates. *Renewable Energy* 2011;36:789–96. <https://doi.org/10.1016/j.renene.2010.08.006>.
- [23] Alam Hossain Mondal M, Sadrul Islam AKM. Potential and viability of grid-connected solar PV system in Bangladesh. *Renewable Energy* 2011;36:1869–74. <https://doi.org/10.1016/j.renene.2010.11.033>.
- [24] RETScreen | Natural Resources Canada n.d. <https://www.nrcan.gc.ca/energy/software-tools/7465> (accessed May 3, 2019).
- [25] Lee KH, Lee DW, Baek NC, Kwon HM, Lee CJ. Preliminary determination of optimal size for renewable energy resources in buildings using RETScreen. *Energy* 2012;47:83–96. <https://doi.org/10.1016/j.energy.2012.08.040>.
- [26] RETScreen Clean Energy Management Software - energypedia.info n.d. https://energypedia.info/wiki/RETScreen_Clean_Energy_Management_Software (accessed May 3, 2019).
- [27] How to Design Solar PV System - Guide for sizing your solar photovoltaic system n.d. http://www.leonics.com/support/article2_12j/articles2_12j_en.php (accessed May 3, 2019).
- [28] Chandel M, Agrawal GD, Mathur S, Mathur A. Techno-economic analysis of solar photovoltaic power plant for garment zone of Jaipur city. *Case Stud Therm Eng* 2014;2:1–7. <https://doi.org/10.1016/j.csite.2013.10.002>.
- [29] Chung HS, Tse KK, Hui SYR, Mok CM, Ho MT. A novel maximum power point tracking technique for solar panels using a SEPIC or Cuk Converter. *IEEE Trans Power Electron* 2003;18(3):717–24.
- [30] Elsayed AT, Mohammed OA. Design control and management of P-V system for residential applications with weak grid connection. 2013. p. 1–10.
- [31] Hussein BN. Introducing a PV design program compatible with Iraq conditions. *Energy Procedia* 2013;36:852–61. <https://doi.org/10.1016/j.egypro.2013.07.098>.
- [32] Oji JO, Idusuyi N, Aliu TO, Petinrin MO, Odejobi OA, Adetunji AR. Utilization of solar energy for power generation in Nigeria. *Int J Energy Eng* 2012;2:54–9. <https://doi.org/10.5923/j.ijee.20120202.07>.
- [33] Electric power transmission and distribution losses (% of output) | Data n.d. <https://data.worldbank.org/indicator/eg.elc.loss.zs> (accessed May 3, 2019).
- [34] SOFO 2018 - The State of the World's Forests, (n.d.). <http://www.fao.org/state-of-forests/en/> (accessed July 2, 2019).